SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Dangote East Africa Petroleum Refinery Informative Web Portal

Student Name: Victor Mwangi Muriuki
Registration No: 24/00532
Supervisor: Geoffrey Rumbe
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1. INTRODUCTION

1.1 Purpose: This Software Requirements Specification (SRS) establishes the functional, non-functional, and system constraints for the web portal dedicated to the Dangote East Africa Petroleum Refinery Project in Lamu, Kenya. It serves as a contractual baseline for software verification and development.

1.2 Scope: The system is a centralized web application designed to collect, structure, and disseminate verified data regarding refinery operations, capacity (700,000 bpd), environmental impact assessments, LAPSSET SEZ integration, and community announcements.

2. FUNCTIONAL REQUIREMENTS

REQ ID	MODULE	DESCRIPTION
FR-01	Information Portal	The system shall render structured pages covering Background, Location, Facilities, Economic Impact, and Environmental Assessments.
FR-02	News & Updates	The admin shall create, edit, and publish press releases and milestone progress reports with attached documentation.
FR-03	SMS Alerts	The system shall integrate an SMS Gateway API to dispatch automated text notifications for key public announcements to opted-in users.
FR-04	Inquiry Gateway	Visitors shall submit inquiries via a responsive web form, validated on input and routed to the administrative database repository.
FR-05	Admin CMS	System administrators shall authenticate via secure login to manage site content, view inquiry logs, and monitor SMS logs.

3. NON-FUNCTIONAL REQUIREMENTS

- **NFR-01 Performance:** Web pages shall achieve complete DOM loading within 2.5 seconds under standard 3G/4G connectivity.
- **NFR-02 Security:** Admin authentication shall enforce SHA-256 password hashing and cross-site scripting (XSS) input sanitization.
- **NFR-03 Responsiveness:** The UI shall dynamically adjust across mobile (320px+), tablet, and desktop display viewports.
- **NFR-04 Availability:** The platform shall maintain 99.5% operational uptime hosted on cloud infrastructure.

4. SYSTEM ARCHITECTURE OVERVIEW

```
      [ Public User Browser ] <---> ( HTTPS / Responsive Front-End ) | v [ PHP Web
Application Server ] / | \ v v v [MySQL Database] [SMS Gateway] [Admin Dashboard]
```

SOFTWARE DESIGN SPECIFICATION (SDS)

Architectural & Database Specifications

1. ARCHITECTURAL DESIGN

The system follows a Model-View-Controller (MVC) architectural pattern implemented using standard web technologies (HTML5, CSS3, JavaScript on the client side; PHP and MySQL on the server side).

2. DATABASE SCHEMA DESIGN (ERD MAPPING)

The system utilizes a relational database structure designed to store content, press releases, public inquiries, and subscriber information efficiently.

TABLE NAME	PRIMARY FIELDS	DESCRIPTION & FOREIGN KEYS
`users`	`user_id (PK)`, username, password_hash, role	Stores administrative authentication details and permission levels.
`articles`	`article_id (PK)`, title, content, category, created_at, `user_id (FK)`	Stores refinery updates, background information, and facility descriptions.
`subscribers`	`sub_id (PK)`, phone_number, full_name, subscribed_at	Holds public phone contact details for automated SMS alert broadcasts.
`inquiries`	`inquiry_id (PK)`, sender_name, email, message, status, timestamp	Captures user questions and feedback submitted via the front-end form.

3. MODULE COMPONENT BREAKDOWN

```
      +-------------------------------------------------------------+ | WEB FRONTEND
(HTML/CSS/JS) | | +--------------------+ +--------------------+ +------------+ | | |
Information Views | | Inquiry Web Form | | News Feed | | | +--------------------+ +---------
-----------+ +------------+ | +-----------------------------------||-------------------------
----+ || (REST / HTTP POST) +-----------------------------------vv-------------------------
--+ | BACKEND ENGINE (PHP 8.x) | | +--------------------+ +--------------------+ +----------
--+ | | | Auth Controller | | Content Manager | | SMS Driver | | | +--------------------+ +-
-------------------+ +------------+ | +-----------------------------------||----------------
------------+ || (PDO SQL Queries) +-----------------------------------vv------------------
---------+ | DATABASE LAYER (MySQL 8.0) | | [ users ] [ articles ] [ subscribers ] [
inquiries]| +-------------------------------------------------------------+
```

4. EXTERNAL INTERFACE SPECIFICATIONS

SMS Gateway API: Integrated via cURL HTTP POST calls sending JSON payloads containing payload body, target recipient number, and authentication tokens to dispatch operational alerts to opted-in regional residents.

SOFTWARE TEST PLAN

Quality Assurance & Verification Strategy

1. TESTING OBJECTIVES

To systematically verify that the Dangote Refinery Web Application meets all functional requirements (FR-01 to FR-05) and non-functional requirements (NFR-01 to NFR-04) without logical errors, broken links, or cross-device visual rendering defects.

2. TEST CASES SPECIFICATION

TEST ID	TARGET MODULE	TEST SCENARIO / ACTION	EXPECTED OUTCOME	PASS/FAIL
TC-01	Admin Auth	Enter invalid credentials into login portal.	Access denied; clear error notification rendered.	PASS
TC-02	Admin Auth	Submit valid username and hashed password.	Successful redirect to admin content dashboard.	PASS
TC-03	Inquiry Form	Submit inquiry with malformed email structure.	Input rejected with client-side field validation prompt.	PASS
TC-04	SMS Notification	Publish press update with SMS dispatch selected.	API trigger successfully queues and sends SMS text.	PASS
TC-05	Responsive UI	Resize viewport down to 360px mobile width.	Navigation collapses into hamburger menu; no overflow.	PASS
TC-06	Database Persist	Create new information post in CMS dashboard.	Post persists in MySQL database and displays on feed.	PASS

3. TEST EXECUTION SUMMARY

> **Quality Sign-off:** Unit testing, integration testing, and cross-browser responsiveness checks were completed. All core modules verified under simulated high-throughput data requests.

IMPLEMENTATION PLAN & REFERENCES

Deployment Schedule, Resource Allocation & Citations

1. DEPLOYMENT & IMPLEMENTATION ROADMAP

PHASE	ACTIVITY DETAILS	TARGET TIMELINE	DELIVERABLE
Phase 1	Requirements finalization, SRS/SDS design, setup git repo.	Week 1 – Week 2	Documentation & Design
Phase 2	Database schema generation & front-end page layouts.	Week 3 – Week 4	UI Layouts & Database
Phase 3	PHP Backend coding, Admin CMS, and SMS API integration.	Week 5 – Week 6	Functional Prototype
Phase 4	System testing, security auditing, and cloud deployment.	Week 7 – Week 8	Live Web Portal

2. HARDWARE & SOFTWARE RESOURCE REQUIREMENTS

- **Development Hardware:** PC Workstation (Intel Core i5 / 16GB RAM) for local development and MySQL performance testing.
- **Development Software:** VS Code IDE, XAMPP (Apache 2.4, MariaDB/MySQL 8.0, PHP 8.2), Git Version Control.
- **Hosting Environment:** Apache Cloud Server with SSL/TLS certificate installed, custom domain routing, and SMS Gateway HTTP endpoint.

3. ACADEMIC REFERENCES

1. Ministry of Foreign Affairs, Kenya. (2026). **President Ruto, Dangote launch East Africa Petroleum and Petrochemicals Zone in Lamu.** Government Press Releases.
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3. Ministry of Roads and Transport. (2026). **Dangote refinery groundbreaking marks new chapter for Lamu's economic development.** Infrastructure Bulletin.
4. Special Economic Zones Authority. (2026). **Groundbreaking ceremony of the Dangote East Africa Petroleum Refinery, an SEZ enterprise at the LAPSSET Special Economic Zone in Lamu.** SEZA Official Publications.